# Ali Can Keskin

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## Professional Summary

Software engineer with 5 years in real-time XR streaming, in C++ and C#. I own the client side of an enterprise streaming product on 9 device families, and I'm turning it from one app per device into a standard OpenXR API layer. I also design the company's OpenXR extensions.

## Experience

**Software Engineer** *Munich, DE*
*Hololight* *Jun 2022 – Present*
*XR streaming software company in Munich, about 70 staff. Team of about 10 engineers.*

- Lead developer of the **Stream** clients on 9 device families: HoloLens 2, Meta Quest, Magic Leap 2, PICO, HTC Vive Focus 3, Snap Spectacles, Windows, Android, and Web.
- Turning the per-device clients into one device-agnostic **OpenXR API layer** (session, hands, controllers, frame override), so a new headset needs far less work.
- Designing the company's **OpenXR extensions** for remote rendering, as a member of the Khronos OpenXR working group.
- Added reprojection and motion estimation to our Monado-based **OpenXR runtime**, and tested it against the OpenXR conformance suite (CTS).
- Lead developer on the **Hub Client**, the app customers use on the headset to sign in, browse, and launch streamed apps. Released for Meta Quest, PICO and HoloLens 2, with live updates over SSE and a release pipeline per vendor.
- Designed the shared **WebRTC** connection layer and native **C++**/JNI code for Android headsets, tuned for low latency.
- Built the browser client from scratch in **JavaScript** with WebSockets and **WebXR**, for zero-install streaming.
- Merged 6 platform build variants into one codebase behind platform abstraction APIs.
- Led the team's move to **AI-assisted development**: built LLM-based PR automation in **Python** and wrote guidelines other teams adopted. Set up **CI/CD** for every platform.

**Unity Developer** *Ankara, TR*
*Simovate* *Sep 2021 – Mar 2022*
*Industrial simulation and XR startup, a subsidiary of Simsoft, Ankara. Team of 4–5 engineers. Left after an offer from Hololight; the German work visa took 3 months.*

- Developed AR/VR applications for industrial training and remote support in **Unity** (C#), with performance-critical modules in **C++**.
- Built a **CNC milling machine simulator** in C++, demoed at Oracle's Industry Innovation Lab.
- Built a custom **mesh manipulation library** for real-time cutting and deformation in VR on standalone headsets.
- Created a mobile **remote maintenance app** with live annotations and data streaming, letting remote experts guide field technicians.

**Android Developer** *Ankara, TR*
*Medication Hour* *May 2020 – Sep 2020*
*Freelance project for a pharmacist. Sole developer.*

- Designed and launched **Medication Hour**, an Android app on Google Play for managing medication schedules with custom reminders.
- Built the **Java** and **Firebase** backend for real-time sync, authentication, and push notifications.
- Owned the full product lifecycle: concept, UI design, development, release, and maintenance.

**Software Engineering Intern** — *Ankara, TR*
*HAVELSAN* — *Jul 2020 – Aug 2020*
*Turkish defense and IT company. Summer internship.*

- Worked on the **payroll module** of **KOVAN**, HAVELSAN's enterprise ERP system.
- Implemented calculation modules, queries, and REST API endpoints in **Java/Spring Boot** to automate payroll processing.

**Game Development Intern** — *Ankara, TR*
*Erik Games* — *Jun 2019 – Jul 2019*
*Indie game studio. Summer internship.*

- Worked on gameplay features and performance for **PlayStation 4** games in **Unreal Engine 4** and **Unity**.
- Contributed to 3 published titles: Brain in Retro Space, Alpha Invasion, Brain Breaker.

## Education

**TED University** — *Ankara, TR*
*B.E. in Computer Engineering* — *Sep 2017 – May 2021*
*Minor in Business Administration*

- GPA: 3.48/4.0
- **Graduation Project:** Virtuanance, a WebRTC-based AR/VR remote maintenance app
- **Relevant Coursework:** Computer Architecture, Computer Vision, Machine Learning, Simulation

## Skills

**Languages:** C++, C#, JavaScript, Java, Python, SQL

**Technologies:** OpenXR (API layers, extensions, runtime), WebRTC, WebSockets, WebXR, REST APIs, Unity, OpenGL, Unreal Engine

**Practices & Tools:** Git, CMake, CI/CD, Unit & Integration Testing, Code Review, JNI

**Platforms:** Android, Windows, Web, XR headsets (Meta Quest, PICO, HTC Vive Focus 3, HoloLens 2, Magic Leap 2, Snap Spectacles)

**Specializations:** Real-Time Systems, Low-Latency Networking, Cross-Platform Architecture, Performance Optimization, AR/VR

## Projects

**Procedural Planets**: *GPU-Accelerated Planet Renderer* — GitHub

- Procedural planet renderer in **C++17/OpenGL 4.3+**. Compute shaders generate terrain in real time with icosahedron-based LOD.
- Features atmospheric scattering, ocean rendering, biome classification, and multi-layer noise (Simplex + fractal).

**contexture**: *AI Coding Setup for Claude Code* — GitHub

- My setup for working with AI in **Claude Code**: coding rules that load before related code, a review pass after, safety hooks, and memory. Built in **Node.js**.

## Certifications

**Machine Learning with Python: From Linear Models to Deep Learning** — *Dec 2020*
*Massachusetts Institute of Technology (MIT)*

- Covered neural networks, reinforcement learning, linear and nonlinear classification, and collaborative filtering. View Certificate